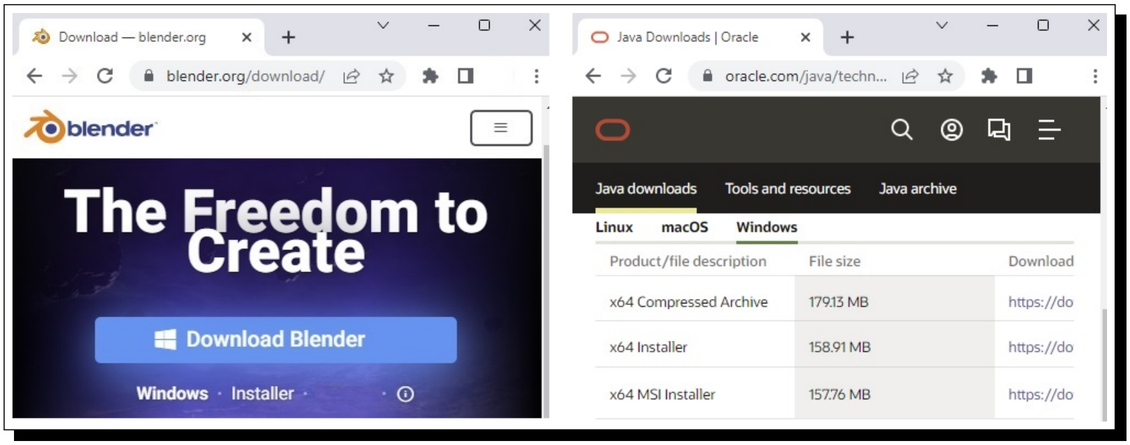
# SOFTWARE INSTALLATION ON WINDOWS

1. **Blender Simulation Software**

   - Go to *Blender Website* and download the latest version of installer.
   - Run the executable installer and follow the instructions for installation.



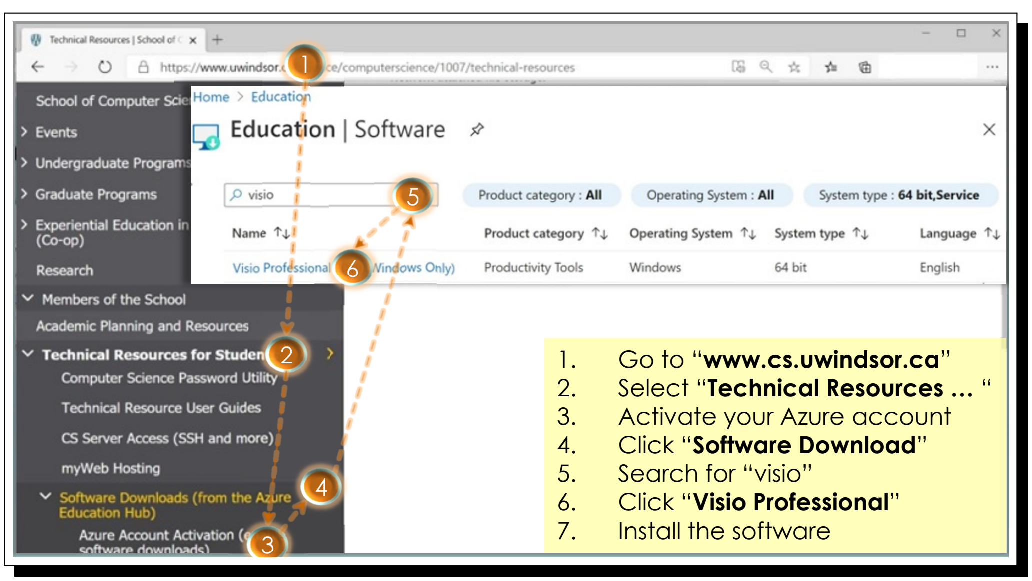

2. **Java Development Kit (JDK)**

   (a) Go to *Oracle Website* and download the latest version of JDK installer.

   (b) Run the executable installer and follow the instructions for installation.

3. (optional) **Visio Professional**

   - Connect to the University network via GlobalProtecct (refer to *this link* for installation)
   - Follow the instructions in the figure below to download and install the latest version of Visio Pro available at the Azure Ecuation Hub.



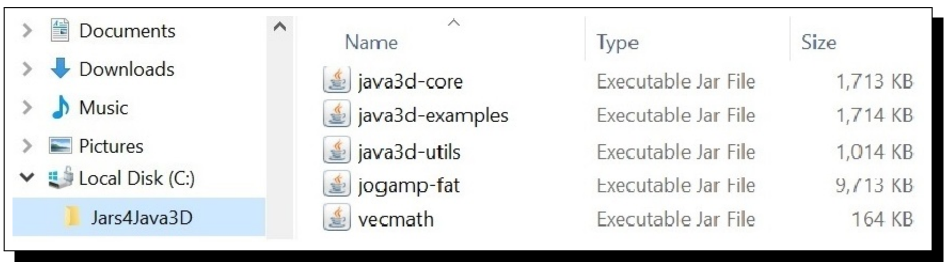

4. **Java 3D Installation**

(a) The Jars for Java3D are retrievable in a zipped file (shown in the above figure) available at *this link* for the latest stable version. The JogAmp fat JAR is also retrievable *here* as `jogamp-fat-all.7z` file. The jar files in both zip files can be extracted with *7-Zip*.

(b) Create a folder under the `C:/` and name it `Jars4Java3D`, i.e., `C:/Jars4Java3D`.

(c) Extract to this folder all the jar files in `jogamp-java3d1.7.1-final.7z`, including `j3d-core.jar`, `j3d-examples.jar`, `j3d-utils.jar`, and `vecmath.jar`.

(d) Extract to the same folder the `jogamp-fat.jar` file in `jogamp-fat-all.7z`.



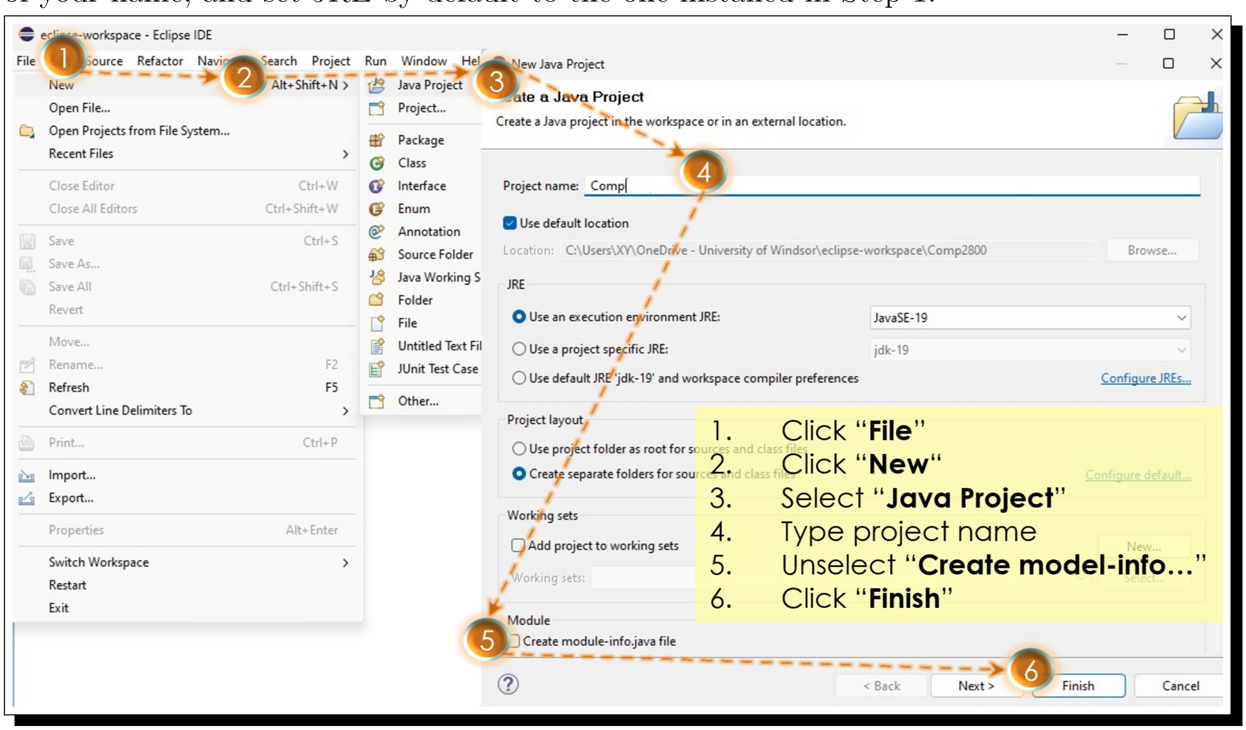


5. **Eclipse Installation and Setup**

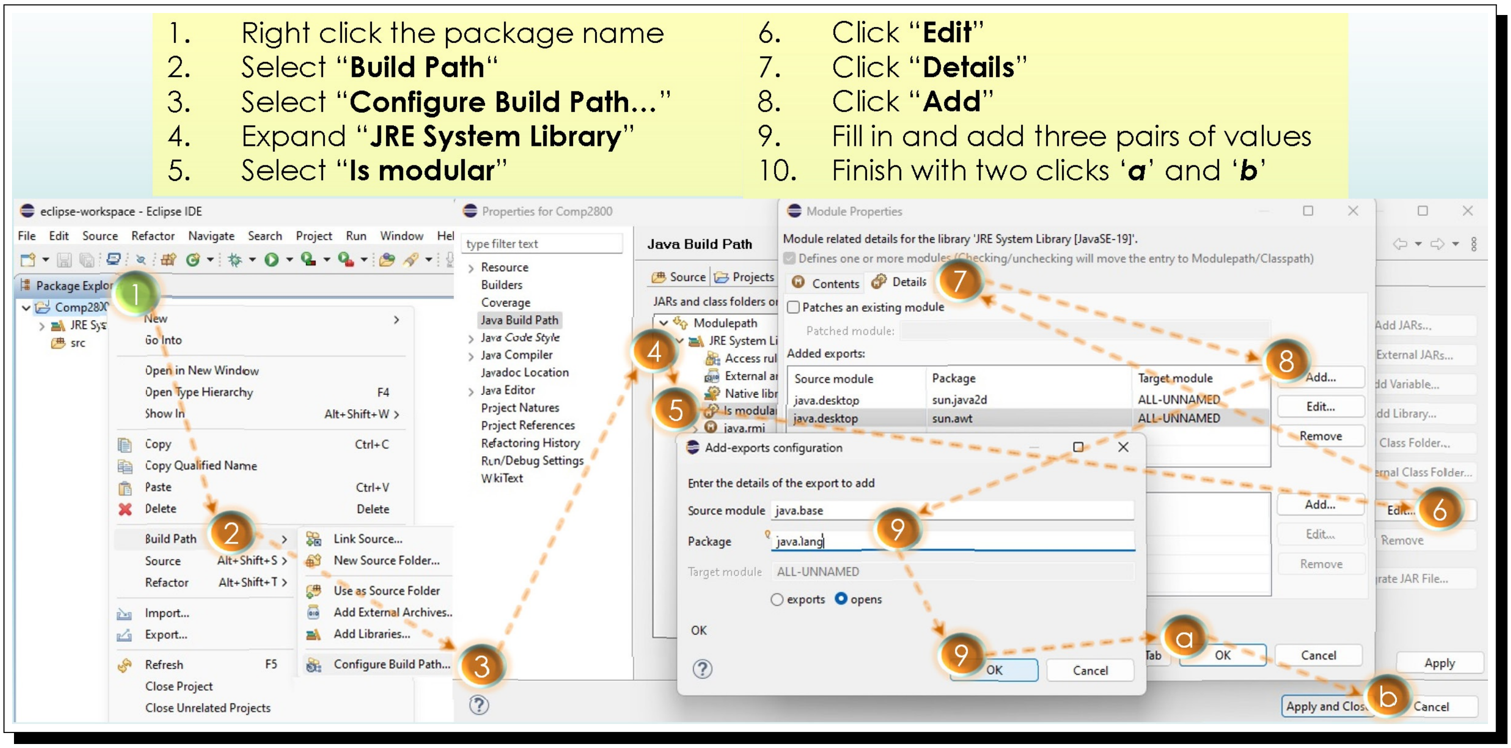
(a) Download the installer from *www.eclipse.org/downloads/* as shown in the above figure.

(b) Start *eclipse* after the installer finishes installing "`Eclipse IDE for Java Developers`".

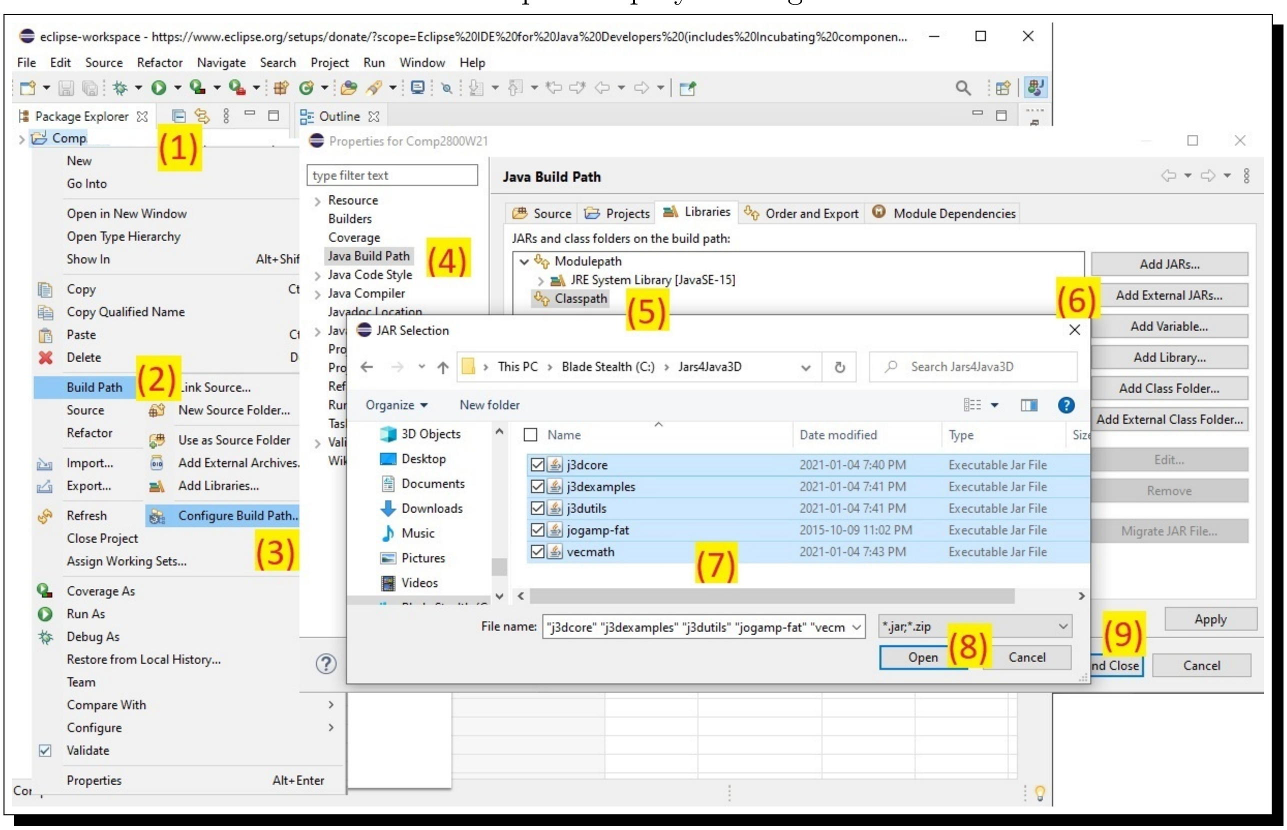
(c) Create a new Java project as `Comp8490XY`, where `XY` needs to be replaced by the initials of your name, and set JRE by default to the one installed in Step 1.



(d) After right clicking the newly created project (e.g., `Comp8490XY`), select `Build Path` and then `Configure Build Path` in the pop up menus. Afterwards, follow the steps to add three VM arguments, which sets `java.base/java.lang`, `java.desktop/sun.awt`, and `java.desktop/sun.java2d` to `ALL-UNNAMED`.




(e) While still in the pop-up window of `Configure Build Path`, select `Classpath` under the `Libraries` tab in the window opened up by clicking `Java Build Path`.



(f) Choose to `Add External JARs` and open the jar files after locating them in the folder. The setup is completed after applying the changes and closing the popup windows.

(g) Create a java class file and name it to `RotatingCube.java`, whose codes can be copied from the file posted in `Resources->Java3D Files` folder.

(h) Installation is successful if `RotatingCube.java` program renders a spinning colored cube as shown in the right figure. Pressing the arrow keys should change the cube's location/size. Pressing the '=' key brings the cube back to its original location.

(i) Online documentation for the Java3D classes is available by following this *link*.